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Ecological research and management of intermittent rivers: an historical review and future directions

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SUMMARY

1. Rivers and streams that do not flow permanently (herein intermittent rivers; IRs) make up a large proportion of the world’s inland waters and are gaining widespread attention. We review the research on IRs from its early focus on natural history through to current application in management and policy.

2. The few early studies of the ecology of IRs were largely descriptive. Nevertheless, in the 1970s, synthesis of this sparse research complemented work on temporary standing waters to found a powerful framework for much of the subsequent research on IRs.

3. Research on the ecology and biogeochemistry of IRs continues to fuel our understanding of resistance and resilience to drying and flooding as disturbances. Syntheses of the growing literature, including cross-continental and cross-climate comparisons, are revealing the generality and individuality of ecological and ecosystem responses to flow cessation and surface water loss. Meanwhile, increasing numbers of experiments test the causality of these responses.

4. Much of the increased consideration of IRs in research, management and policy is driven by the observed and projected shifts in flow regimes from perennial to intermittent associated with changes in land and water use and climate, superimposed on the high incidence of natural intermittency. The need to protect and better manage IRs is prompting researchers to develop new or modified methods to monitor flow status and assess the ecological condition of these systems.

5. Intermittent river research and management will benefit from greater exploration of aquatic–terrestrial linkages, wet–dry cycling and temporal dynamics, more-detailed mapping and predictive modelling of flow intermittency and the application of metapopulation and metacommunity concepts alongside multiple-stressors and novel-ecosystems research. By building on existing knowledge, continuing to develop quantitative models and distribution maps and using experiments to test hypotheses and concepts, we can further ecological understanding and wise management of these ubiquitous ecosystems.

Keywords: biogeochemistry, ephemeral stream, fish, invertebrate, temporary river

Introduction

Rivers and streams that cease surface flow span a wide range of hydrological regimes and have inspired a correspondingly rich lexicon. They are variously described as non-perennial, temporary, seasonally intermittent, episodic or ephemeral, among other terms (Uys & O'Keeffe, 1997; Steward *et al.*, 2012). The channels of these systems may dry partially or completely during flow cessation, but it is the phenomenon of flow cessation itself that distinguishes them from perennial rivers and defines them as a group. For simplicity, we refer to these systems as intermittent rivers (IRs).

Intermittent rivers are found throughout the world yet our understanding of their ecology and how best to manage them has trailed behind that of perennial systems (Datry, Larned & Tockner, 2014b). There are signs, however, that IR research may be 'catching up' with research on perennial systems. Research on IRs is rapidly increasing in scope and depth of insight (Datry, Arscott & Sabater, 2011a) and the recent appearance of IR-based conceptual papers and opinion pieces (e.g. Larned *et al.*, 2010b; Sponseller, Heffernan & Fisher, 2013; Datry *et al.*, 2014b), cross-system comparisons (e.g. Datry *et al.*, 2014a) and studies addressing the challenges of IR assessment, protection and restoration (e.g. Acuña *et al.*, 2014; Arthington, Bernardo & Ilhéu, 2014; Datry *et al.*, 2014b) all indicate a shift in research focus and growing interest in IR ecology and management. Many factors drive this shift, from the observed and projected transitions in river flow regimes from perennial to intermittent (and vice versa) associated with climate, land- and water-use changes (e.g. Postel, 2000; Reich *et al.*, 2010; Döll & Schmied, 2012) to adjustments in regional water policies to improve IR management and resolve confusion over legislative protection status (e.g. Nadeau & Rains, 2007; Nikolaidis *et al.*, 2013). Furthermore, special sessions at international symposia and journal issues on IR research (e.g. *Aquatic Sciences*, 2011, 73(4); *River Research and Applications*, 2014, 30(10); *Freshwater Biology*, this issue) have stimulated and consolidated research networks and projects dedicated to IR research and management (e.g. Datry *et al.*, 2014b; Prat *et al.*, 2014).

Our aim in this study was to review the literature on IR ecology and management to explore the development and expansion of IR research over time, synthesise key findings and highlight major advances in knowledge and understanding. We focus on several key topics: early research themes and natural history; invertebrates and fish; biogeochemistry; hydrological and ecological

assessment; conservation and restoration; and policy. While these topics do not include all those in the extensive literature on IRs, they provide a targeted yet broad coverage of scientific knowledge of IR ecology and management that falls within the scope of our expertise. For four of the topics (invertebrates, fish, biogeochemistry and assessment), we also use bibliometric approaches (see Tables S1 & S2) to quantitatively illustrate the geographical expansion of IR research through time and reveal trends in development and parallels among topics.

Early research and natural history

The few early papers on the ecology of IRs focussed almost solely on the natural history of their aquatic inhabitants and were broadly descriptive (e.g. Allee & Torvik, 1927; Stehr & Branson, 1938). Variation in the physical, chemical and biological characteristics of IRs was typically described in much the same way as for perennial systems, and the importance of intermittency to aquatic fauna was not emphasised. Nonetheless, one of the earliest IR studies (Allee & Torvik, 1927) described the spatial distribution of both aquatic and terrestrial fauna (invertebrates, fish, amphibians and lizards) in isolated pools and on the dry bed of an IR, and several papers specifically outlined changes in the natural history of stream biota during drying (e.g. Moon, 1956). Researchers became intrigued by the continuing presence of aquatic fauna in IRs despite periodic drying. For instance, IRs were initially viewed as harsh environments for fish, many of which were assumed intolerant of flow intermittency and surface-water loss (Williams & Coad, 1979). This may also partly explain the smaller number of studies of fish than invertebrates in the IR literature (Table S1). Gradually, an awareness of post-drying recolonisation mechanisms (e.g. Larimore, Childers & Heckrotte, 1959) and the importance of life cycles and life-history traits (John, 1964; Clifford, 1966) emerged. However, most early studies of IRs had a narrow focus on one aspect of a single IR and, with a few exceptions (e.g. Clifford, 1966; Harrel & Dorris, 1968), did not attempt to explore hypotheses about the general ecology of IRs.

It was not until the late 1970s that a series of landmark papers synthesised the handful of published studies of IRs to explicitly derive some 'general remarks' (Williams & Hynes, 1976, 1977; Williams, 1977; Williams & Coad, 1979). Williams & Hynes (1977) proposed that the fauna of IRs consisted of three main groups:

permanent stream species that can survive short dry periods, facultative species that occur in both lotic and lentic waters and specialised species that are adapted to non-permanent waters. They built on earlier work by proposing (i) several strategies by which invertebrate species survived the dry period (as aerial adults, by reinvasion from nearby waters, buried in sediments, under rocks, under leaf litter or in other instream refuges such as crayfish burrows and remnant pools), (ii) that these strategies reflected the species' life cycles and life-history traits (e.g. timing of egg hatching, rapid-growth phases and emergence as adults) and (iii) that variations in life cycles among species created faunal successions as IRs transitioned from flowing streams to isolated pools to dry channels. These 'general remarks' formed the platform for much subsequent work on IRs as research became increasingly predictive, based on conceptual models that emphasised the driving force of water regime.

During this period, research on temporary standing waters stimulated the development of IR ecology. Up to the 1980s, the ecology of temporary standing waters was far better researched than that of IRs, possibly because habitats like ponds were more tractable for sampling than streams and the taxonomy and biology of their fauna were better known. As most IRs comprise temporary standing waters for part of the time, it was understandable that ecologists studying IRs would draw on this literature to supplement their research. One of the most influential publications was the review by Wiggins, Mackay & Smith (1980), which comprehensively synthesised work on the ecological and evolutionary strategies of the biota of temporary pools. There was also growing recognition that ecosystem processes, such as organic matter breakdown, differed between temporary and permanent standing waters. Pioneering work by Bärlocher, Mackay & Wiggins (1978) showed that decaying detritus exposed to air during the waterless period had higher protein content and palatability to aquatic invertebrates than detritus submerged in nearby permanent waters. This was hypothesised to be partly responsible for the rich animal communities of temporary standing waters and led to experimental tests of this idea in IRs (e.g. Boulton, 1991).

By the late 1980s, ecological research on IRs was extending beyond its focus on natural history and across a widening array of countries and, by extension, climate zones (Figs 1, S1 & S2). As the literature base continued to expand through the 1990s and into the 21st century (Fig. 2), several influential books and reviews were published (Williams, 1996, 2006; Gasith & Resh, 1999;

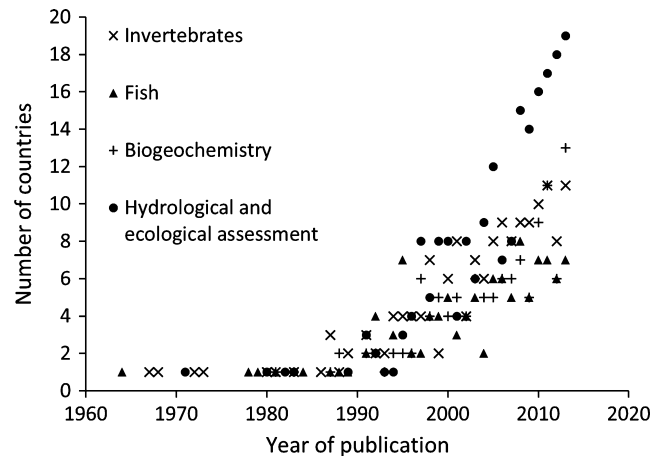


Fig. 1 The number of countries in which intermittent river (IR) research is conducted has increased over time for each of the four topics of IR research on which we conducted literature searches: invertebrates, fish, biogeochemistry and hydrological and ecological assessment. Publications by topic span a total of 37, 30, 27 and 49 countries, respectively. See Tables S1 and S2 for detailed information on the searches. Data are derived only from studies conducted in one country only and exclude the year 2014 because it was not complete at the time of the searches (21 August 2014).

Boulton, 2003; Lake, 2003; Magoulick & Kobza, 2003; Larned *et al.*, 2010b) and IR research became increasingly characterised by studies considering theoretical aspects (e.g. disturbance and ecological succession; Table 1), along with the assessment, protection and restoration of IR habitats and networks. Today, IRs in both xeric and mesic regions are recognised as playing essential roles in controlling material fluxes and maintaining biodiversity (Datry *et al.*, 2014b) and providing important habitat for multiple migratory and locally endemic species (Labbe & Fausch, 2000; Courtwright & May, 2013; Kerezszy *et al.*, 2013).

Invertebrates and fish

Early research on natural history of IRs set the foundation for future synthesis and conceptualisation, which primarily began in the late 1970s, continuing into the 1980–90s. Studies of invertebrates and fish in IRs in this formative period initially took somewhat different directions, with research on fish considering landscapes as filters alongside the stability and persistence of communities (e.g. Smith & Powell, 1971; Meffe & Minckley, 1987) and invertebrate research conducting faunal comparisons with nearby perennial sites and systems (e.g. L  gier & Talin, 1973). In particular, insights from the synthetic studies of IRs and their invertebrate communities in Canada were extended to other regions, including Australia and the U.S.A, revealing the need

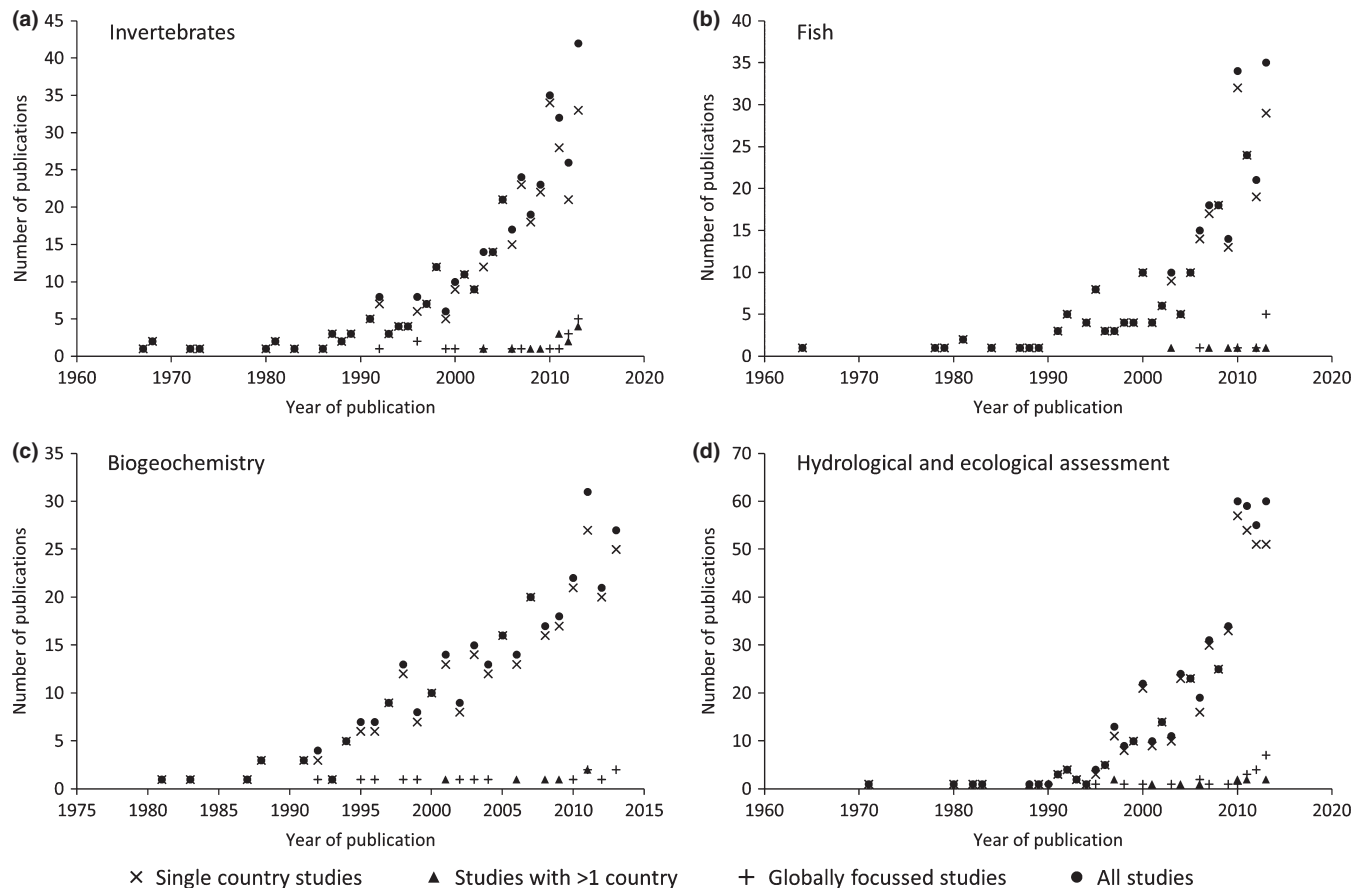


Fig. 2 The number of publications has increased over time for each of the four topics of intermittent river research on which we conducted literature searches: (a) invertebrates, (b) fish, (c) biogeochemistry and (d) hydrological and ecological assessment. See Tables S1 & S2 for detailed information on the searches. Data exclude the year 2014 because it was not complete at the time of the searches (21 August 2014).

for more-detailed cross-continental comparisons to better understand faunal responses to drying as a disturbance. For example, considerable overlap was found between the invertebrate faunas of permanent streams and IRs in Australia, which contrasted with results from Canada (Boulton & Suter, 1986), yet there were remarkably similar taxa across instream refuges in IRs of both countries (Boulton, 1989). Later work, however, showed that this similarity did not extend to all systems. For example, instream refuges in Sycamore Creek, an IR in the Sonoran Desert, U.S.A., contributed little to the resistance of the invertebrate communities to drying (Boulton *et al.*, 1992). Instead, recolonisation from nearby perennial sites was the primary mechanism driving community persistence in the intermittent habitats (Stanley *et al.*, 1994).

Several years later, an influential cross-continental IR study by Boulton (2003) compared responses of invertebrate communities to stream drying between Australia and the U.K. This study focussed attention on the 'critical stages' of stream drying, specifically that changes in invertebrate community richness, abundance and trophic

organisation followed a 'stepped response': gradual change up to a critical threshold (e.g. flow cessation) followed by a swift transition when habitats disappeared or fragmented and water quality declined. Additionally, the study showed that although the invertebrate communities of IRs in both countries had rapid rates of recolonisation post-drying, the recolonisation of taxa without desiccant-resistant stages or high mobility was likely to be delayed. Boulton's (2003) conceptual model was recently extended by Bogan, Boersma & Lytle (2015) to include post-drying trajectories of recovery, likewise highlighting a stepped pattern but one dependent on the relative resistance and resilience of invertebrate taxa rather than on abiotic changes in habitat. These models may also have some application to fish. For example, fish communities can be structured by local abiotic conditions such as declining water quality and pool size in drying pools of IRs (Capone & Kushlan, 1991; Magoulick, 2000). Hydrological variation and extremes in flow in IRs, including floods and droughts, also influence spatiotemporal patterns of fish community

Table 1 Key ecological theories and conceptual models developed from or extended to intermittent rivers, with selected examples

Theory or model	Key finding, influence, application or concept	Examples
Ecological succession	Succession theory received considerable empirical and theoretical support from studies of IRs, often conducted during flow periods or in perennial sections embedded within IR networks	Fisher <i>et al.</i> (1982), Grimm & Fisher (1989)
Flood Pulse Concept	The extension of this concept to dryland rivers, including many IRs, showed how their 'boom-bust' ecology relates to high flow variability at multiple temporal scales	Walker <i>et al.</i> (1995)
Telescoping Ecosystem Model	An ecosystem-based conceptual model describing how disturbances such as drying reset and alter ecosystem processes such as nutrient spiralling	Fisher <i>et al.</i> (1998b)
Disturbance ecology	IRs provide useful environments in which to study the resistance and resilience of biota to both flooding and drying. Such research inspired the influential 'stepped response' model of invertebrate community responses to drying, applied not only to IRs but also to perennial systems undergoing novel drought conditions	Resh <i>et al.</i> (1988), Stanley <i>et al.</i> (1994), Boulton (2003), Davey & Kelly (2007), Datry <i>et al.</i> (2014a), Bogan <i>et al.</i> (2015)
Hyporheic Refuge Hypothesis	Tested in IRs to see whether the hyporheic zone is a refuge from drying and/or flooding, with equivocal results	Del Rosario & Resh (2000), Datry (2012), Stubbington (2012)
Punctuated biogeochemical reactors	Proposes that wetting and drying cycles govern most ecosystem processes in IRs	Larned <i>et al.</i> (2010b)
Coupled aquatic–terrestrial ecosystems	IRs incorporate spatial and temporal alternations of dry and wet phases within channels as well as lateral linkages across riparian zones and floodplains	Larned <i>et al.</i> (2010b), Datry <i>et al.</i> (2016)
Metapopulations, metacommunities and metaecosystems	'Meta' concepts can be readily tested in IRs because their disturbance regimes generate habitat fragmentation at different scales of time and space	Larned <i>et al.</i> (2010b), Robson <i>et al.</i> (2011), Falke <i>et al.</i> (2012)
Reconciliation ecology	A realistic way of managing highly modified IRs, as opposed to restoring ecological states that may not be sustainable or achievable	Moyle (2014)

structure (e.g. Balcombe & Arthington, 2009; Matthews *et al.*, 2013), with reductions in flow magnitude and permanence often associated with declines in species abundances and richness (e.g. Davey & Kelly, 2007).

Concepts of disturbance were also explored in some of the first studies on hyporheic fauna in IRs (e.g. Cooling & Boulton, 1993), including those examining whether the hyporheic zone was used as a refuge by surficial invertebrates during drying (e.g. Del Rosario & Resh, 2000; Datry, 2012; Table 1). Despite decades of research, evidence that surficial invertebrates use hyporheic zones during drying is equivocal (Stubbington, 2012) and research on this topic continues (e.g. Storey & Quinn, 2013). Recent work suggests the hyporheic zone can play an important role in maintaining populations of fish in IRs (Kawanishi *et al.*, 2013). Ultimately, the hyporheic zone should be viewed as one important component in the broad suite of refuges that facilitate community resistance and resilience to drying (Stubbington, 2012).

While IR research on fish historically focussed on drying perhaps more so than flooding, much research on

invertebrates in IRs centred on floods, with several studies specifically investigating responses to these events from the perspective of temporal succession (e.g. Fisher *et al.*, 1982; Grimm & Fisher, 1989). Indeed, we know far less about invertebrate responses to drying than to flooding (Lake, 2003), especially in terms of temporal fluctuations. For instance, short-term studies indicate that many invertebrate communities of IRs are temporally unstable in abundance and taxonomic composition (e.g. Bêche & Resh, 2007; Blanchette & Pearson, 2013). Alternating wet and dry phases in IRs, coupled with variation in antecedent flow conditions, contribute further to the high temporal variation (e.g. Sponseller *et al.*, 2010; Datry, 2012; Leigh, 2013). Long-term studies, however, suggest that the biological trait composition of invertebrate communities is much more stable in time than taxonomic composition and abundance (Bêche, McElravy & Resh, 2006). In contrast, taxonomic composition and abundance of fish in IRs may remain relatively stable and persistent in the long term despite extreme flooding or drying (e.g. Meffe & Minckley, 1987; Matthews *et al.*, 2013).

The survival of both fish and invertebrates in IRs is influenced by their biological and behavioural traits, particularly in relation to life-history strategies and dispersal capacities. For example, some fish have extended spawning or multiple cohorts, which may compensate for unpredictable or short periods of water presence and habitat connection in IRs, while others spawn during periods of no flow, maintaining populations in isolated pools of IRs between the sometimes brief periods of hydrological connection (Alkins-Koo, 2000; Paugy, 2002; Kerezszy *et al.*, 2011). Fish in some IRs disperse over greater distances than their perennial stream counterparts, a possible counter measure against negative effects of habitat loss and isolation on population dynamics (Walker & Adams, 2014). However, local extinctions and population bottlenecks have been linked with intermittency and habitat contraction (e.g. Sousa *et al.*, 2010; Huey *et al.*, 2011) and significant genetic structuring in fish populations of highly fragmented and isolated IR habitats indicates that not all IR fish species are strong dispersers, with some displaying limited dispersal even over short distances (Bartáková *et al.*, 2013). Recent work on invertebrates suggests dispersal strength is a key determinant of metapopulation and metacommunity structure in highly fragmented IR networks, emphasising the importance of perennial patches of habitat for ensuring population viability (Cañedo-Argüelles *et al.*, 2015). The quantity and quality of available refuges, such as nearby permanent pools and perennial reaches, play critical roles in the resistance and resilience of invertebrates and fish to drying (Pires, Cowx & Coelho, 1999; Magoulick & Kobza, 2003; Perry & Bond, 2009; Beesley & Prince, 2010; Robson *et al.*, 2013; Walker, Adams & Adams, 2013). For example, residual pools in IR networks allow some fish to survive dry periods or habitat isolation, simply due to the sustained presence of water in otherwise dry channels (e.g. May & Lee, 2004; Wigington *et al.*, 2006) and/or the absence of competitors or predators (e.g. Erman & Hawthorne, 1976; Kerezszy *et al.*, 2013).

One of the first studies to compare the effects of both high flows and drying on fauna between intermittent and perennial systems (Miller & Golladay, 1996) suggested that the relative importance of resistance and resilience to disturbance differed between them, with long-term assemblage stability being maintained by taxa that were either highly resistant (in IRs) or resilient (in perennial streams). Further investigations found evidence supporting the importance of either resistance, resilience or both mechanisms in underpinning responses to and recovery from drying in IRs (Fritz &

Dodds, 2004; Ledger *et al.*, 2012; Stubbington & Datry, 2013; Boersma *et al.*, 2014; Datry *et al.*, 2014a). However, practically all studies agree that prolonged and consecutive flow-cessation events beyond durations 'normally' experienced will lower the capacity of most fauna to persist in IRs, regardless of whether that persistence is driven by resistance or resilience (e.g. Robson, Chester & Austin, 2011; Hershkovitz & Gasith, 2013; Matthews *et al.*, 2013). Accordingly, research is increasingly focussed on understanding impacts of climate change, human-induced flow modifications of naturally intermittent systems and water diversions that induce intermittency on population persistence and community composition (e.g. Balcombe *et al.*, 2011; Alexandre, Ferreira & Almeida, 2013).

Recent interest in the ecological roles of terrestrial fauna in IRs has broadened the research field beyond a purely aquatic perspective, for example by highlighting the diversity and distinctness of invertebrate communities inhabiting the surfaces of dry riverbeds (e.g. Wisheart, 2000; Steward *et al.*, 2011; Corti, Larned & Datry, 2013). High levels of omnivory and reciprocal transfers of energy between the terrestrial and aquatic compartments of IRs help to buffer food webs against extreme flow variation. Extensions of such work on invertebrates (Leigh *et al.*, 2010) to fish (Sánchez-Carmona *et al.*, 2012; Courtwright & May, 2013; Wesner, 2013; Blanchette *et al.*, 2014) and terrestrial vertebrates (Leigh, Reis & Sheldon, 2013b) provide growing support for the concept of IRs as coupled aquatic-terrestrial ecosystems (*sensu* Larned *et al.*, 2010b; Table 1), whereby hydrological variability shifts mosaics of aquatic and terrestrial habitat within and across IR channels, riparian zones and floodplains, influencing ecosystem responses. For example, research in IRs has been instrumental in generating the now widely accepted view that aquatic, autotrophic microbial sources of carbon (e.g. algae) can be as important as terrestrial, detrital sources (e.g. leaf litter) in fuelling aquatic food webs, particularly during dry phases following retreat of surface waters to isolated pools (e.g. Closs & Lake, 1994; Bunn, Davies & Winning, 2003; Reid *et al.*, 2008).

Biogeochemistry

A series of novel experiments in the surface, hyporheic and parafluvial zones of Sycamore Creek and other Sonoran Desert IRs in the 1980–90s (e.g. Grimm, Fisher & Minckley, 1981; Grimm, 1987; Holmes, Fisher & Grimm, 1994; Jones, Fisher & Grimm, 1995) contributed to a growing interest in biogeochemical processes in rivers,

both perennial and intermittent. Much of this effort followed the Stream Solute Workshop (1990) organised by the North American Benthological Society to synthesise conceptual and empirical developments in stream-tracer studies. Nitrate dynamics were particularly well studied in Sycamore Creek, although focus tended to remain on flood pulses, bed instability and hyporheic flow paths, rather than intermittency or stream drying *per se* (although see Stanley & Boulton, 1995). The first explicit examination of the relationships between intermittency and storage of organic matter and nutrients in Sycamore Creek did not appear until a decade later (e.g. Sponseller & Fisher, 2006). Nevertheless, the early studies in Sycamore Creek spawned a novel concept in freshwater ecology and perhaps the first explicitly developed in IRs: the telescoping ecosystem model (TEM; Fisher *et al.*, 1998b; Table 1). The TEM considers stream-corridor systems as nested cylindrical elements with processing lengths that extend and retract in response to disturbance regimes, being one of the first ecosystem-based concepts to consider the roles of flooding and drying in rivers as linked hierarchical landscapes (see also Fisher *et al.*, 1998a, 2001). Another striking output of the research on desert IRs such as Sycamore Creek was the convergence of aquatic and terrestrial perspectives in nutrient biogeochemistry (Grimm *et al.*, 2003), leading to recognition of IRs as coupled aquatic–terrestrial ecosystems (Larned *et al.*, 2010b; Datry *et al.*, 2014b; Table 1).

Research now suggests that sediments in IRs and the times during and immediately after dry phases are potential biogeochemical ‘hot spots’ and ‘hot moments’ (McClain *et al.*, 2003; Amalfitano *et al.*, 2008; Gallo *et al.*, 2014). Consequently, processes occurring during dry, wet and transitional phases are increasingly being considered in biogeochemical models (e.g. Butturini, Bernal & Sabater, 2005; Tzoraki *et al.*, 2009) and empirical studies (e.g. McIntyre *et al.*, 2009; Datry *et al.*, 2011b; Timoner *et al.*, 2012). For example, research has revealed the preconditioning effects of sunlight during dry phases on leaf-litter decomposition and the subsequent bioavailability of organic carbon (e.g. Dieter *et al.*, 2011; Fellman, Petrone & Grierson, 2013). The ‘first pulses’ of flow and organic debris marking the end of dry periods (e.g. Corti & Datry, 2012) can trigger hypoxic blackwater events in which large quantities of freshly mobilised dissolved organic carbon are rapidly metabolised (e.g. Hladysz *et al.*, 2011).

These studies led to the view of these systems as ‘punctuated biogeochemical reactors’ (Larned *et al.*, 2010b; Table 1), with nutrients and organic matter being stored, processed or transported in response to spatio-

temporal fluctuations in drying and wetting (e.g. Corti *et al.*, 2011; Kerr *et al.*, 2011). An important implication of this view is that budgets and large-scale estimates of nutrient and organic matter fluxes in rivers (e.g. Battin *et al.*, 2009; Acuña & Tockner, 2010; Raymond *et al.*, 2013) are likely to be inaccurate if IRs are excluded from analyses or the spatiotemporal dynamics of water, solutes or organic material (live and detrital) are overlooked (Kerr *et al.*, 2011; Datry *et al.*, 2014b). Mesocosm studies and field experiments further show the importance of dry phases and IRs in contributing to the overall input and metabolism of organic material in river systems (e.g. Gerull *et al.*, 2011; Hladysz *et al.*, 2011). Such insights will enable us to better understand biogeochemical processes in IRs that contribute to water quality or involve hazardous substances and how to mitigate their impacts (e.g. Ademollo *et al.*, 2011; Tzoraki *et al.*, 2014). For instance, well-documented remediation measures such as riparian reforestation have been proposed to be as, if not more, important in IRs as in perennial systems for reducing nutrient and heavy metal loadings because the episodic and intense runoff events characteristic of many IR systems intensify catchment influences and land-degradation impacts (e.g. Leigh *et al.*, 2013a; Tzoraki *et al.*, 2014).

Hydrological and ecological assessment

Assessments of flow status in IRs initially attempted to identify water availability and flood risk in the absence of widespread, continuous gauge data (e.g. Kisiel, Duckstein & Fogel, 1971). Over time, greater availability of field-derived flow-permanence data and technical advances in statistics, GIS and remote sensing have allowed researchers to better assess existing cartographic databases, map IRs, monitor flow intermittency and inform decisions related to best management practices or jurisdictional determinations (e.g. Heine, Lant & Sengupta, 2004; Jaeger & Olden, 2012; Lang *et al.*, 2012; Callow & Boggs, 2013).

Today, a wide variety of data sets, techniques and spatial scales are used to assess and map flow status and hydrological variation in IRs (e.g. Johnson *et al.*, 2009; Fritz, Wenerick & Kostich, 2013b; Fritz *et al.*, 2013a). Researchers continue to produce useful classifications of IR flow regimes (e.g. Poff, 1996; Uys & O’Keeffe, 1997; Kennard *et al.*, 2010) and regionalisations of their distributions (e.g. Snelder *et al.*, 2013). Tools have also been developed to help managers implement or maintain ecological flows in IRs in the face of extensive land degradation and growing demands to divert or withhold

water for human use. Larned *et al.* (2010a), for example, present a four-step framework for analysing longitudinal flow variation in rivers, quantifying both flow magnitude and state (flowing or dry) and exploring its ecological consequences. Alternatively, in the absence of gauges, measures of channel geomorphology may provide a proxy for long-term changes in IR discharge, for example in response to changes in catchment land use (Costelloe, Grayson & McMahon, 2006).

Research on the ecological condition assessment of IRs also has increased in complexity and inferential power. Much of the early work consisted of case studies documenting impacts of silviculture and livestock agriculture on the hydrology, water quality and sediment runoff from IRs and their catchments (e.g. Douglass & Van Lear, 1983; Neary, Bush & Grant, 1986) or the ecological consequences of mining contaminants and sewage or industrial effluent discharges (e.g. Lewis & Burraychak, 1979; Coimbra, Graça & Cortes, 1996). These studies were followed by a growing awareness and assessment of the consequences of effluent-dominated flows (e.g. Hassan & Egozi, 2001; Brooks, Riley & Taylor, 2006), water resource development (e.g. Leigh & Sheldon, 2008) and climate change (e.g. Lawrence *et al.*, 2010) on IR flow regimes and ecological condition, with an emphasis on the importance of maintaining natural flow variability for biodiversity conservation and ecosystem functioning.

As researchers began to investigate the applicability of ecological assessment tools initially developed for perennial rivers, including bioassessment indicators (e.g. Morais *et al.*, 2004) and water quality models (e.g. Mannina & Viviani, 2010), to IRs, the challenges and constraints of assessing the condition of systems having high levels of hydrologic variation across space and time became apparent. Studies continue to highlight the need for new or modified assessment methods (Boulton *et al.*, 2000; Sheldon, 2005; Ademollo *et al.*, 2011; Dallas, 2013). Empirical bioassessment studies, particularly from southern Europe but also from Australia and parts of the U.S.A. where intermittent channels dominate river networks, also address the challenge of determining reference conditions in IRs (e.g. Prat *et al.*, 2014). Most of these studies focus on characterising intersite and temporal variation across reference-site classifications (e.g. Lunde *et al.*, 2013; Feio *et al.*, 2014), following Sheldon (2005). However, few studies have measured condition across gradients of one or more stressors (e.g. Sánchez-Montoya, Vidal-Abarca & Suárez, 2010), identified stressor-specific indicators or designated water quality criteria for IRs (e.g. Sheldon & Fellows, 2010). Promising avenues to address the challenges of assessing 'stream health' in systems where

water is not consistently present in space or time include the use of microbial biomarkers (Wilkes *et al.*, 2013), hyporheic invertebrates (Leigh *et al.*, 2013c) and the terrestrial fauna of dry riverbeds (Steward *et al.*, 2011) as bioindicators of ecological condition, while continuing to investigate when and where existing measures and thresholds developed for perennial systems apply to IRs (e.g. Mazor *et al.*, 2014; Prat *et al.*, 2014).

Conservation and restoration

Many IRs have suffered degradation from their use as rubbish disposal sites and drains for wastewater and sewage effluent, or from impacts of catchment vegetation clearing, urbanisation, mineral extraction and flow modification (Larned *et al.*, 2010b; Steward *et al.*, 2012). Widespread recognition of the conservation significance of IRs has been slow and remains inadequate (Acuña *et al.*, 2014), despite scientific evidence since at least the late 1990s that IRs help maintain biodiversity in the landscape (e.g. Kingsford, Boulton & Puckridge, 1998; Katz, Denslow & Stromberg, 2012) and perform fundamental ecosystem services such as nutrient cycling (Bernhardt & Palmer, 2011) and groundwater renewal (e.g. Sophocleous, 2002).

As the importance of the hydrological variability of IRs to their ecology was recognised (e.g. Poff & Ward, 1989; Walker, Sheldon & Puckridge, 1995), scientists raised concerns that this variability was threatened by flow regulation for irrigation and other human activities, especially in dryland regions (Davies *et al.*, 1994; Walker, Puckridge & Blanch, 1997; Boulton *et al.*, 2000). As a result, initial efforts to conserve IRs focussed on protecting their natural flow variability (O'Keeffe, 1989; Kingsford *et al.*, 1998). Over time, conservation efforts broadened to include identifying and protecting refuge pools for aquatic fauna during dry phases (e.g. Sheldon *et al.*, 2010; Kadye & Chakona, 2012; Hermoso, Ward & Kennard, 2013) and maintaining connectivity to adjacent ecosystems so that metapopulations and ecosystem processes persist at the landscape scale (e.g. Kerezszy *et al.*, 2013; Dexter *et al.*, 2014). Scientists have also argued for recognising the value of IRs with extremely ephemeral flows; usually dry, they are especially prone to being dismissed in environmental impact assessments, and limited knowledge about their ecosystem services hinders their conservation (Boulton, 2014).

Comprehensive programs focused at catchment-scale conservation of IRs are rare. One of the earliest programs was initiated on the San Pedro River, Arizona in 1988 when the San Pedro River Nature Conservancy

Area was established to protect over 50 km of the riparian zone of the upper river basin. Ecologists documented the success of this protection, identifying the hydrological regimes and groundwater interactions necessary for conserving keystone species (e.g. Lite & Stromberg, 2005) and developed assessment programs to monitor the effects of groundwater withdrawals beyond the conservation area (Stromberg *et al.*, 2006). The general applicability of insights gained from such programs to other IRs remains largely unexplored.

Gradually, there has also been acknowledgement that IRs damaged by human activities, in their catchments or the channels themselves, may need active restoration (e.g. http://www.albarine.com/actions_realisees.html, <http://www.arroyoseco.org/watershed.htm>). However, most river restoration programs focus on perennial systems (e.g. Palmer & Hondula, 2014), reflecting the time lag in science and management applied to non-perennial systems. Along with the general lack of documentation and monitoring of restoration projects (Palmer *et al.*, 2007), the main challenge facing successful restoration of IRs is probably that of overcoming prevailing Western cultural preferences for perennial flows (Boulton *et al.*, 2000) and stable meandering channels (Kondolf, Podolak & Grantham, 2013). Indeed, many IRs have been made artificially perennial, for example through the continual input of wastewaters or via regulation (e.g. Reich *et al.*, 2010).

In IRs whose flow is now regulated for irrigation and other human needs, there have been a few attempts, often short-term or experimental, to recover pre-regulation hydrological variability by strategically timed environmental releases of water from upstream dams (Stromberg *et al.*, 2007; Witze, 2014). However, the success of these releases in IRs has often been hampered by an inability to match natural hydrological and sediment regimes (Jacobson & Jacobson, 2013) as well as constraints imposed by legislated obligations to meet the needs of downstream human communities. Restoring natural periods of intermittency is even less likely to be appreciated by most human users because potential effects (e.g. proliferation of exotic species and loss of some species of large fish targeted by anglers; Bond *et al.*, 2010; Reich *et al.*, 2010) could be perceived as negative. In IRs, restoring hydrological variability, especially intermittency, poses challenges of implementation (e.g. Kondolf *et al.*, 2013) and requires courage (Boulton *et al.*, 2000) and legislative support.

In most cases, flow regimes, catchments and the water quality of IRs have been so altered, along with invasion by exotic species, that novel ecosystems have been cre-

ated whose restoration back to natural conditions is virtually impossible. Moyle (2014) suggests that 'reconciliation ecology', which acknowledges that humans increasingly dominate most ecosystems on the planet (Rosenzweig, 2003), is a more feasible goal and he gives several examples where it could be applied to IRs in California. This is likely to be the future direction of restoration programs in IRs elsewhere, enabling establishment of realistic policy goals to help manage river flows and infrastructure and achieve ecological targets.

The slow but growing recognition in policy of the prevalence and importance of IRs

As our scientific understanding of IRs has developed, the importance of including them in policy and management decisions has been increasingly recognised. The legal status and definitions of IRs, and the extent to which they are incorporated into such decisions, vary both within and among countries and create much debate (Acuña *et al.*, 2014). Human-induced shifts in flow regimes from perennial to intermittent coupled with the high incidence of natural intermittency, prolonged droughts and increasing demand for fresh water have all increased the need to define and consider IRs and their conservation importance in water policy and planning (Porto & Porto, 2002; Nadeau & Rains, 2007; Shinker *et al.*, 2010; Acuña *et al.*, 2014).

In the U.S.A., IRs received considerable scientific attention after the Supreme Court decision on *Rapanos versus United States* in 2006, which created challenges for identifying the physical boundaries for implementation of the Clean Water Act (e.g. Nadeau & Rains, 2007; Leibowitz *et al.*, 2008; Caruso, 2011). Research on IRs at that time was considered limited in its ability to inform policymakers regarding these challenges and specific attention to IRs was lacking in science-forum papers, explicitly written to inject science into relevant policy. A boom in IR research productivity ensued (Figs 2 and S1) as researchers and managers in the U.S.A. became increasingly concerned with demonstrating the ecological links between non-navigable, often intermittent, channels and downstream navigable waters. In particular, calls were made for accurate mapping of IRs, predictive models of the spatiotemporal extent of intermittency, large-scale studies and models to quantify upland IR influences on downstream systems and greater understanding of the capacities of IRs to remove or transport harmful materials (e.g. pollutants) and act as refuges (e.g. for migratory fish during extreme conditions) (Leibowitz *et al.*, 2008; Acuña *et al.*, 2014; U.S. EPA, 2015).

In Europe, although implementation of the Water Framework Directive (WFD) influenced the direction of much research towards river condition assessment, the legal and therefore protection status of IRs still very much depends on how 'waterbodies' are classed by authorities in each river basin district (Munné & Prat, 2004). To help address this, the European MIRAGE project was developed to provide scientific knowledge to better assess the ecological status of IRs (focussed on the Mediterranean region) and better understand their ecological functioning for improved flood management (Nikolaidis *et al.*, 2013). The project both stimulated and relied on hydrological and ecological IR research to support recommendations to the WFD of explicit IR definitions and adaptive assessment methods to suit the different hydrological conditions that IRs exhibit through space and time (Nikolaidis *et al.*, 2013; Prat *et al.*, 2014).

In Australia, possibly because of the wide recognition of the prevalence of these systems across the landscape, IRs are explicitly included in many river basin management plans and recognised by most legislative definitions (Acuña *et al.*, 2014). The challenges presented to decision-makers and managers by IRs that cross political boundaries have also received scientific attention (Kingsford *et al.*, 1998).

Yet despite the growing consideration of IRs in policy, many calls in the scientific literature for improved policy on water provision for the environment in IRs, and development of integrated water resource management (IWRM) policies for IR basins (e.g. O'Connor, 2001), studies centrally concerned with policy that affects IRs (e.g. Thoms & Sheldon, 2002) remain scarce. In some regions, societal and cultural constraints and financial or technical limitations may make the implementation of globalised ideals of IWRM for IRs difficult or inappropriate (Swatuk & Motsholapheko, 2008). This may be particularly relevant for developing countries, although IWRM policies in IR basins in these regions are increasingly being developed and supported by scientific understanding and modern software tools (e.g. Benito *et al.*, 2010; Mounir *et al.*, 2013). The low esteem in which IRs are held in some regions may also impede environmentally sound policy implementation and management (Saurí-Pujol *et al.*, 2001; Armstrong *et al.*, 2012).

The future of IR research and management

The history of IR research has shown its numerous important contributions to scientific concepts and developments and to river management (Table 1) and high-

lighted an emerging focus on experiments and integrating knowledge across multiple studies to explain patterns and processes. Several 'hot topics' of research are also emerging, including application of metapopulation and metacommunity concepts; exploration of aquatic-terrestrial linkages and temporal dynamics in IRs; IR mapping and predictive modelling of flow intermittency; and multiple-stressors and novel-ecosystems research (e.g. Davis, Sim & Chambers, 2010; Cañedo-Argüelles *et al.*, 2015; Datry *et al.*, 2016). We believe the exploration of these topics will be essential to improve understanding and management of IRs.

First, mechanistic and predictive understanding of ecological responses to flow intermittency and surface-water loss is vital for the continued scientific development of IR research. To improve this understanding, we must increasingly supplement knowledge based on single systems and short-term studies conducted within single hydrological phases (i.e. flowing (aquatic), drying (transitional) or dry (terrestrial) phases alone) with cross-system and cross-phase comparisons, multi-scale approaches (e.g. Labbe & Fausch, 2000; Lowe, Likens & Power, 2006) and long-term (>10 years) research (e.g. <http://nature.berkeley.edu/reslab/>; <http://caplter.asu.edu/>). Cross-phase and long-term research will be crucial to capture and understand lagged ecological responses to past drying and flooding in IRs and better predict future responses (e.g. Sponseller *et al.*, 2010).

Second, piecemeal management of river sections may limit the attainment of desired targets (e.g. maintaining populations of endangered species) because perennial and non-perennial components of river ecosystems interact through time and space in complex ways. Unfortunately, even in regions where IRs predominate, disproportionately more attention still tends to be given to documenting the extent and condition of perennial systems. Improved mapping of IRs, particularly headwater streams and over large spatial scales, and models for predicting flow permanence in IRs are needed to improve estimates of water and material fluxes within and across landscapes (e.g. Moraetis *et al.*, 2010; Raymond *et al.*, 2013; Snelder *et al.*, 2013), implement water-resources policies and manage IRs as networks.

Third, we need improved understanding and predictive capacity about how IRs might respond to anthropogenic stressors and future changes. There is a pressing need to understand how water resource development, land use and climate change will interact with intermittency and affect biogeochemical processes, the persistence of extant populations and communities, trophic organisation and the potential for species invasions in IRs (e.g.

Puckridge, Walker & Costelloe, 2000; Bogan & Lytle, 2011; Ledger *et al.*, 2013; Rolls *et al.*, 2013). Many IRs, both natural and artificial, are affected by multiple pressures (e.g. chemical, geomorphological, hydrological) and are likely to experience novel conditions along with complex stressor interactions that may alter ecological patterns and processes in unexpected ways. Applying multiple-stressors and/or novel-ecosystems research to IRs (e.g. Davis *et al.*, 2010) has much to offer.

One factor that has hindered IR research is the deeply engrained idea that IRs primarily occur in arid, semi-arid and Mediterranean-climate regions. The consideration of IR research beyond these regions and increasing confirmation that IRs occur across all major terrestrial biomes and climates (e.g. Stanley, Fisher & Grimm, 1997; Figs 1 & S1) are slowly helping to change this view. However, accurate spatial data on distributions of IRs across biomes are needed to consolidate overdue change in perceptions. More consistent terminology (e.g. via standardised key words) and explicit identification of study systems as IRs will further help, simplifying the collation and synthesis of IR research. Intermittency is not a uniform feature of a few aberrant systems but a dynamic spectrum seen across many streams and rivers worldwide. To sustainably manage our rivers to support native biodiversity while maximising human benefits, we must build on the existing IR knowledge base, continue to develop quantitative models and distribution maps and use more experiments to test hypotheses and concepts. In doing so, we will enhance our ecological understanding and wise management of these globally ubiquitous ecosystems.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Fig. S1. The increasing geographical and climatic scope of IR research on invertebrates (I), fish (F), biogeochemistry (BGC) and hydrological and ecological assessment (HEA) through time, as indicated by the results of literature searches as detailed in the main article, and in Tables S1 and S2 (excluding 'global' studies).

Fig. S2. Proportional distribution of IR research across countries by topic: a invertebrate communities, b fish, c biogeochemistry, d hydrological and ecological assessment (as indicated by the results of literature searches detailed in the main article, and Tables S1 and S2).

Table. S1. Literature searches used to review IR research within and across topics.

Table. S2. Selected publications from Web of Science searches associated with the four research topics identified in the main article

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